

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To develop an AI-powered Smart Lender system that predicts whether a loan applicant is eligible for a loan using machine learning models, helping financial institutions make faster and more accurate lending decisions.
Scope	The project includes data collection, preprocessing, feature engineering, model training, evaluation, and deployment of a web application that predicts loan approval based on applicant details.
Problem Statement	
Description	Traditional loan approval processes are slow, manual, and may lead to inaccurate decisions. Financial institutions need an intelligent system that can quickly evaluate loan applications with better accuracy.

Impact	The system reduces loan processing time, improves prediction accuracy, minimizes financial risk, and enhances customer satisfaction through faster approvals.
Proposed Solution	
Approach	The Smart Lender system uses machine learning algorithms such as Decision Tree, Random Forest, KNN, and XGBoost to analyze applicant data and predict loan approval. The best-performing model is integrated into a Flask web application for real-time predictions.
Key Features	- AI-based loan eligibility prediction. - Multiple machine learning model comparison. - Real-time loan approval prediction.



	User-friendly Flask web application. Fast and accurate decision-making.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		

Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv